

CAPSTONE PROJECT

Resume Screening and Role Matching using NLP

Enterprise AI-Powered Applicant Tracking System (ATS)

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Project Type: **Data Science with GenAI Capstone Project**

Company: **TalentBridge Solutions Pvt. Ltd.**

Tech Stack: Python | NLP | Machine Learning | Streamlit | Scikit-learn



Business Problem & Objectives

Current Recruitment Challenges

- Thousands of resumes for every job opening
- Manual screening consumes recruiter time
- Keyword matching misses good candidates
- Inconsistent recruiter decisions
- High recruitment cost
- Increased hiring time

Project Solution

- Automate resume screening
- Reduce recruiter workload
- Improve candidate-role matching
- Generate ATS-style Fit Scores
- Rank candidates automatically
- Build recruiter dashboard

Methodology & Architecture

Our solution employs a robust, multi-stage methodology to ensure accurate and efficient resume screening and role matching.

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1. Resume Dataset

Utilized a comprehensive dataset of 5,000 diverse professional profiles for training and evaluation.

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2. Data Validation & Cleaning

Implemented rigorous data validation and cleaning procedures to ensure data quality, consistency, and eliminate noise from the raw resume data.

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3. Feature Engineering

Developed 7 critical engineered features to capture various dimensions of candidate suitability:

- Resume Quality Score
- ATS Readiness Score
- Technical Strength Score
- Leadership Score
- Domain Alignment Score
- Project Relevance Score
- Candidate Quality Index

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4. NLP Pipeline

Leveraged advanced Natural Language Processing (NLP) techniques, including TF-IDF vectorization and Cosine Similarity, to semantically analyze resume content and job descriptions.

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5. Hybrid ATS Scoring System

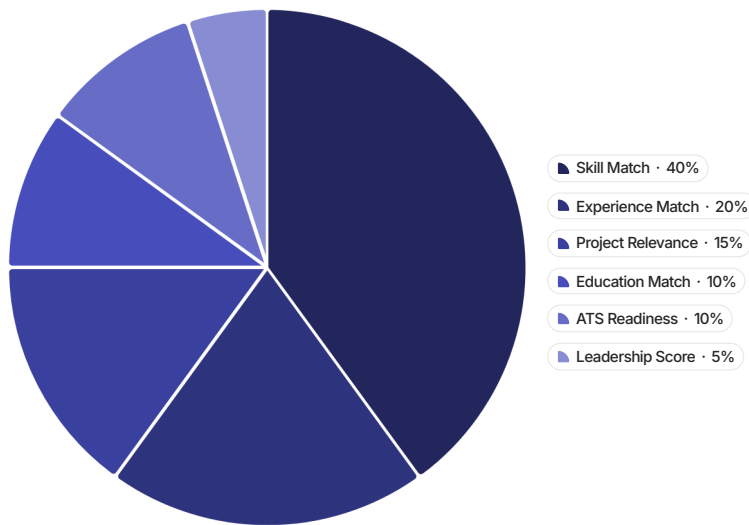
Designed a hybrid scoring model that combines the engineered features with NLP-driven similarity metrics to generate a comprehensive fit score for each applicant.

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6. Candidate Ranking & Dashboard

Developed an intuitive recruiter dashboard that displays ranked candidates, detailed fit scores, and key insights to facilitate informed hiring decisions.

Hybrid ATS Scoring Model



- ❏ Skill Match is the largest contributor at 40%, while the remaining factors create a balanced, explainable overall fit score.

This hybrid approach combines multiple business factors for balanced, explainable recommendations rather than relying solely on keyword matching.

- **Most important:** Skill Match and Experience Match drive 60% of the score.
- **Balanced evaluation:** Project relevance, education, ATS readiness, and leadership add depth.
- **Professional output:** The model supports transparent, recruiter-friendly decisions.

Results & Business Impact

5000

Resumes Validated

Large-scale screening coverage

15+

Business Features

Engineered to support the
workflow

2

Impact Areas

Achievements and business
benefits

Conclusion & Future Scope

Key Takeaways

This project demonstrates how Natural Language Processing and business-driven feature engineering can automate resume screening and improve recruiter decision-making. The system combines structured data analysis, NLP similarity, and Hybrid ATS Scoring to build a practical enterprise recruitment solution.

- ❏ From screening efficiency to smarter decision support, the solution is designed for real-world recruiting workflows.

Future Enhancements



Resume PDF Parsing



Real Job Descriptions Integration



BERT Embeddings & Transformers



Large Language Models (LLMs)



Explainable AI (XAI)



Cloud Deployment



Recruiter Feedback Learning Loop

Thank You | Questions?